

metal enclosure, which also has to be conductive (no painting/coating) with un-painted overlaps. Fig. 9 shows cross-sections of an enclosure for better understanding.

Electrostatic discharge

Electrostatic discharge occurs when 10-15kV charges accumulated on a person or object get discharged through the electronic product. The ESD damages the semiconductors permanently. Its effect is high when the atmosphere has less humidity, but it can never be avoided. Unless protection measures are taken, the product can get damaged very easily and silently. There are two ways to protect a product from ESD:

1. Provide ESD protection at inputs of the semiconductors that are exposed to the outside world. These inputs are typically connectors, which calls for circuit design with proper ESD protection. The protection devices are called transient voltage suppressors. These reduce the ESD to a safe level, preventing

damage to the semiconductors.

2. Ensure the enclosure has no gaps through which ESD can sneak and the enclosure is grounded properly to act as an excellent ESD shield. While metal enclosures have a good ESD shielding capacity, plastic cabinets are not so lucky. Irony is plastic cabinets are also poor in protection against EMC/EMI and shielding internally is critical for the safety of a product.

Normally, good EMC/EMI countermeasures should work for ESD too in most cases. However, improper grounding in the mains supply can sometimes make the product a victim to ESD. A successful method to prevent PCBAs from getting damaged by ESD is to provide copper trace of 1-2mm width around the edge of the PCB. (The trace end should not be connected to form a loop; it should be open and one end should be connected to chassis ground.) Other candidates that attract ESD are the exposed connector pins (especially berg sticks). The in-circuit program-

ming connectors for the microcontroller programming normally are the culprits. These measures help in ESD voltages being diverted to the grounded trace instead of damaging the semiconductors.

Most of the times ESD solution can be implemented only after the testing of the product has been done after taking all precautions during its designing. ESD standards, however, do allow a system to restart after the start in some applications. But restarts are not possible in mission-critical domains like medical, avionics, and some industrial equipment. **EFY**

The author has 35 years of design experience for various domains and global customers. This article is based on the last chapter in his book titled 'Introduction to Electronic Packaging – Unconventional Guide to Product Design.'



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